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1. Introduction

For this report, a retreat case study has been chosen to arrange a three-day retreat for all management staff from the three campuses of OPAIC. The objective of this report is to perform the analysis of project management methodologies that may be practical to a retreat for OPAIC management staff and determine and justify which methodology is the most suitable for this project. This project demands scheduling a 3-day retreat in term 5 for management employees in Queens town to discuss the strategic plan for the next five years. Additionally, separate planning sessions and plenary meetings need to be in the retreat, including best practices in Iwi/Māori incorporation. Moreover, hosting a formal dinner with speeches from crucial OPAIC leadership to close the conference on the second evening. Lastly, in the current COVID-19 situation, this project may face various challenges such as being held on schedule, traveling conditions, number of attendees, etc.

The current report is divided into three sections. The first part evaluates three project management methodologies suitable to the OPAIC 3-day retreat project. In the second part, the most effective methodology for this retreat project will be raised and assessed. Three methodologies, namely, Waterfall, Critical path method, and Event chain methodology, have been proposed to proceed with this project. Out of these, Event chain methodology is the most appropriate methodology to perform the retreat project. Lastly, Māori values befitting the project are going to discuss.

2. Project Management Methodologies (A 3-day Retreat)

2.1. Waterfall

The first methodology identified for the OPAIC three-day retreat project is Waterfall. The Waterfall Methodology is one of the earliest and most well-known traditional project management methodologies (Balaji & Murugaiyan, 2012). Andrei et al. (2019) state that the technique of Waterfall enables a project to be divided into various fixed stages, with each step necessitating the evaluation and work from the previous stage.

Figure 1 (Balaji & Murugaiyan, 2012) presents the process of building a project applying the waterfall method. The typical steps to reach a Waterfall project are requirements collections, analysis, design, development, testing, implementation, and maintenance. The first phase is to collect demands for this retreat and analyze the project needs. This three-day trip must include separate sessions, plenary sessions, and a formal dinner. The second phase is a scheduling phase that needs the project manager to design a run sheet for this retreat based on the requirements and analysis in the last stage. This step could cause additional issues during the development and testing phases if this step is skipped. When it comes to the third phase, the development phase, the project manager needs to solve all the questions, refine the

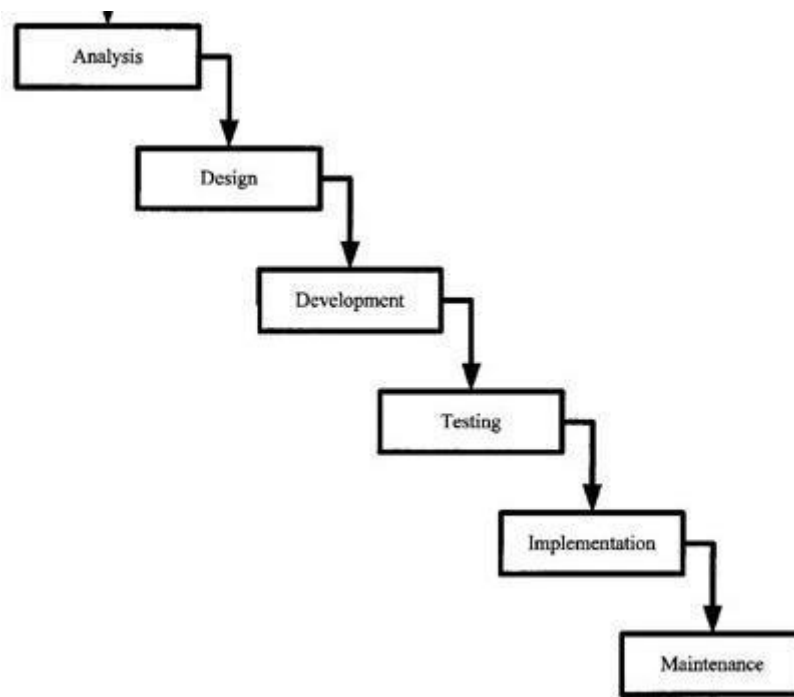


Figure 1 Waterfall Methodology (Balaji & Murugaiyan, 2012)

the resolution and perform each of the requirements in the first phase, applying the run sheet and blueprints from the second phase. Also, practice in Iwi/Māori incorporation should add into consideration. Phase four is testing for a retreat-required conference and a formal part. The venue of this project must be confirmed with enough meeting rooms and a suitable boardroom format. In the fifth phase, the whole project comes to the implementation stage. Each step outlines the tasks that must be completed within a specific time frame. Every activity of this three-day retreat should be advanced and completed on time following the run sheet. Maintenance is the last phase in the Waterfall project.

Waterfall methodology is acknowledged for its precise requirements, ease of implementation, use, and management, and better short-term projects (Eason, 2016). However, Balaji & Murugaiyan (2012) state that each stage is frozen before the next step in the waterfall model. The bright side is that with the help of the Waterfall, all activities can proceed as scheduled and drive the project to success. All the conference participants felt delighted and full of gains, including the Māori management staff. However, there are always some unexpected situations in a retreat, especially in the current COVID-19 condition. Waterfall doesn't allow changing the requirement if the project already comes to the implementation stage. In brief, Waterfall can be a potential methodology for the OPAIC retreat project.

2.2. Critical Path Method (CPM)

The second methodology proposed for the three-day retreat project is the Critical Path Method (CPM). The view of critical path derived from the significant defense projects in the 1950s, a project's dependent

phases that take the least amount of time to accomplish (Musarandega et al., 2020). Reh (2019) stated that CPM is a project methodology that involves identifying each required step and evaluating how long each will take. The objective of CPM is to avoid the types of bottlenecks that afflict a complicated plan. The vital technique for implementing CPM is to build a project model including a checklist of all events needed to accomplish the project, the duration of each event, and the interdependence between the events (Santiago & Magallon, 2009).

When adopting the CPM into a project, it needs to do CPM calculation to obtain its longest path of scheduled activities to completion. Each activity's earliest and latest can begin and end without lengthening the project. Then, the project manager can appoint which events are crucial on the longest path. Santiago & Magallon (2009) also point out that pruning essential events of the path, "Fast-tracking" and "Crashing the critical path," trim the scheduled vital path of a project and give precedence to events for efficient management.

Based on the CPM, for this three-day retreat project, the first phase is to divide the project into three operations required for completion and decide the operational sequence relationship of each activity (Santiago & Magallon, 2009). On day one, the small group meeting needs to be organized, which is the mark of commencement. Only the management staff has the separate sessions; they can bring the ideas to join the plenary conferences. For phase two, CPM requires estimating the time needed for each operation. The whole group numbers should have arrived before 11:00 am on day one, the earliest possible start time for the entire project, and leave at 10:00 am on day three. Determine each activity's earliest possible start/finish time. In phase three, the project manager needs to build time-cost relations and scheduling variations. If the guest check-in 11:00 am on the first day, which is a standard start, and checks out at 10:00 am on the last day. The program can be completed in a planned time with the least cost. The results of are the project will have the least time and higher cost to achieving success. Using CPM on phare three needs to review back to the second phase; the project manager double-checks the missed relationships and prioritizes the critical activities to drive the project to success. Thus, CPM can be a suitable methodology for the OPAIC retreat project.

2.3. Event chain methodology (ECM)

The third methodology proposed for the OPAIC retreat project is the Event chain methodology (ECM). A series of risk events can often alter a project schedule, especially in the current COVID-19 situation. ECM is newly added in project management methodologies. Still, it is not a radically new approach as it is on the basis of available quantitative methods such as Monte Carlo simulation (Avlijaš & Radunović, 2019). According to Virine and Trumper (2016), ECM is based upon six major principles, which are 1) single-

event handling, 2) focus on a series of related occurrences or event chains, 3) establish guidelines for visualizing events or event chains, 4&5) cope with the analysis of the schedule with event chains and 6) explain how to monitor project performance using events or event chains. ECM is a method for analyzing scheduling networks that allow for the modeling and visualization of risk events and is concerned with identifying and assessing risk events as well as situations in which one event triggers another (Avlijaš & Radunović, 2019).

Virine and Trumper (2016) state that applying ECM to project scheduling and analysis includes five steps. Accordingly, this section will adopt these steps to schedule a three-day retreat project and how ECM drives the project to success. Firstly, build a project schedule model using best-case scenario estimates of duration, expenses, and other factors. The project requires a 3-day high-quality retreat in Queens town with budget constraints which means it will be better to choose weekdays instead of weekends and advance booking. Secondly, create a list of events and event chains, together with their probability and consequences for activities, resources, lags, and calendars. To establish a better opportunity and conditions for OPAIC management staff to discuss the next five-year strategic plan, separate sessions and plenary conferences are critical during this retreat. Individual group sessions should be arranged before the plenary meeting, which facilitates the entire management team to conclude a better development strategy for OPAIC. The duration is limited to 3 days and a formal dinner held on the second evening. Both separate discussion and plenary discussion should be finished before the second day evening. Therefore, the first day afternoon will have small group sessions, while plenary sessions will be scheduled on the daytime of the next day.

Thirdly, adopt Monte Carlo simulations to perform a quantitative analysis. Through Monte Carlo simulations, it is likely to decide a particular time or cost for each activity. With the help of the quantitative analysis, a project manager can present percentiles of start and finish times of each activity on a schedule. In step four, conduct a sensitivity analysis which aids in the identification of critical activities, critical events, and event chains as part of the quantitative analysis. The primary project parameters are influenced by determining activities, important events, and event chains. In this retreat project, the separate and plenary session and a dinner party are the three essential activities: mutual influence and interconnection. The fifth step requires running through the analysis regularly again when conducting a project. With these data, the project manager can reassess risk probability and influence. In a nutshell, EMC can be a potential methodology for the OPAIC retreat project.

3. Critical evaluation of chosen methodology (Event chain methodology)

From the three discussed methodologies above, the Event chain methodology could be recommended to OPAIC as the most suitable methodology for the 3-day retreat project. Unlike the traditional methodology,

ECM recommends applying additional factors to decrease the impact of optional perception and overconfidence, which are the most influential root reasons for bad project time scheduling (Avlijaš & Radunović, 2019). ECM provides flexibility to make amendments to the retreat arrangement. COVID-19 still affects people's daily life, especially the emergence of the Omicron variant. The authorities' government can enact a travel restriction policy that will impact or even disrupt the date of the project. With ECM, the project manager can generate a new project schedule using the updated information to ensure the task is completed successfully.

Moreover, ECM greatly streamlines the definition and analysis of complicated issues consorted with project planning, such as the balance of resources or relevance of events (Virine & Trumper, 2016). Other activities can be triggered by one activity, either immediately or after a delay. The extension or postponement of the breakout sessions will directly affect the timing of the entire group sessions in this retreat. The delay can be seen as an event subscription property in the ECM. Applying the ECM to the retreat offers mitigation efforts to deal with this kind of delay and promptly redirects resources, including the Māori incorporation, to ensure project success. Therefore, the Event chain methodology is the most appropriate for the OPAIC retreat project.

4. Māori inclusion in the project

OPAIC is an education institute owned by local Iwi and is guided by Māori values, which should also apply to this retreat project. In traditional and modern Māori society, Whanaungatanga defines the individual's responsibility and role within the collective group and encourages a sense of togetherness, belonging, and relatedness (Walker, 2003). For this three-day project, Whanaungatanga value is implemented by providing opportunities for discussion for each participant, Māori, and non-Māori. Applying Whanaungatanga in the retreat can build a closer relationship with employees of each campus and ensure the next five years' strategic plan made for the benefit of one campus does not undermine the well-being of the other campus involved.

Manaakitanga is another essential Māori value for the three-day retreat. According to Careerforce (2017), Manaakitanga is the act of looking after someone's mana by expressing friendliness, goodness, hospitality, and showing support for others. When selecting and setting up the retreat venue, it is necessary to make the management staff feel valued and welcome using Māori decorations and indicators. Behaving in retreat ways make each participant feel valued, respected, influential, and accepted. For example, the formal diner serves several Māori cuisine on the second night or a greeting in Māori before the start of each meeting. Consequently, Whanaungatanga and Manaakitanga are suitable for the retreat and contribute to the project's success.

5. References

- Andrei, B., Casu-Pop, A., Gheorghe, S., & Boiangiu, C. (2019). A STUDY ON USING WATERFALL AND AGILE METHODS IN SOFTWARE PROJECT MANAGEMENT. *Journal of Information Systems & Operations Management*, 125-135. <https://www.proquest.com/scholarly-journals/study-on-using-waterfall-agile-methods-software/docview/2237828314/se-2?accountid=39660>
- Avlijaš, G. & Radunović, M., 2019. Application of Event Chain Methodology in Schedule Risk Analysis. *European Project Management Journal*, 9(2), 26-34.
- Balaji, S., & Murugaiyan, M. S. (2012). Waterfall vs. V-Model vs. Agile: A comparative study on SDLC. *International Journal of Information Technology and Business Management*, 2(1), 26-30.
- Careerforce. (2017). Learning Guide Māori operating principles and values. *Careerforce*. (1.1), 14-26.
- Eason, O. K. (2016). Information systems development methodologies transitions: an analysis of Waterfall to agile methodology.
- Musarandega, R., Robinson, J., Priti, D. S., Hakobyan, A., Mushavi, A., Mahomva, A., & Woelk, G. (2020). Using the critical path method to rollout and optimise new PMTCT guidelines to eliminate mother-to-child transmission of HIV in Zimbabwe: a descriptive analysis. *BMC Health Services Research*, 20, 1-11. <http://dx.doi.org/10.1186/s12913-020-05900-4>
- Reh, J.F., (2019, September 2). *How to Use the Critical Path Method to Manage a Project*. The Balance Careers. <https://www.thebalancecareers.com/critical-path-description-and-overview-2276120>
- Santiago, J., & Magallon, D. (2009). Critical path method. *CEE320, Winter 2013*.
- Virine, L., & Trumper, M. (2016). Event chain methodology in details. *Retrieved May, 10, 2016*.
- Walker, D. (2003), "Te ao Maori/ a Maori worldview," in Walker, D., Bunt, P. and Stephens, M. (Eds), *A Cultural Impact Assessment of Nelson Sewerage and Ponds*.